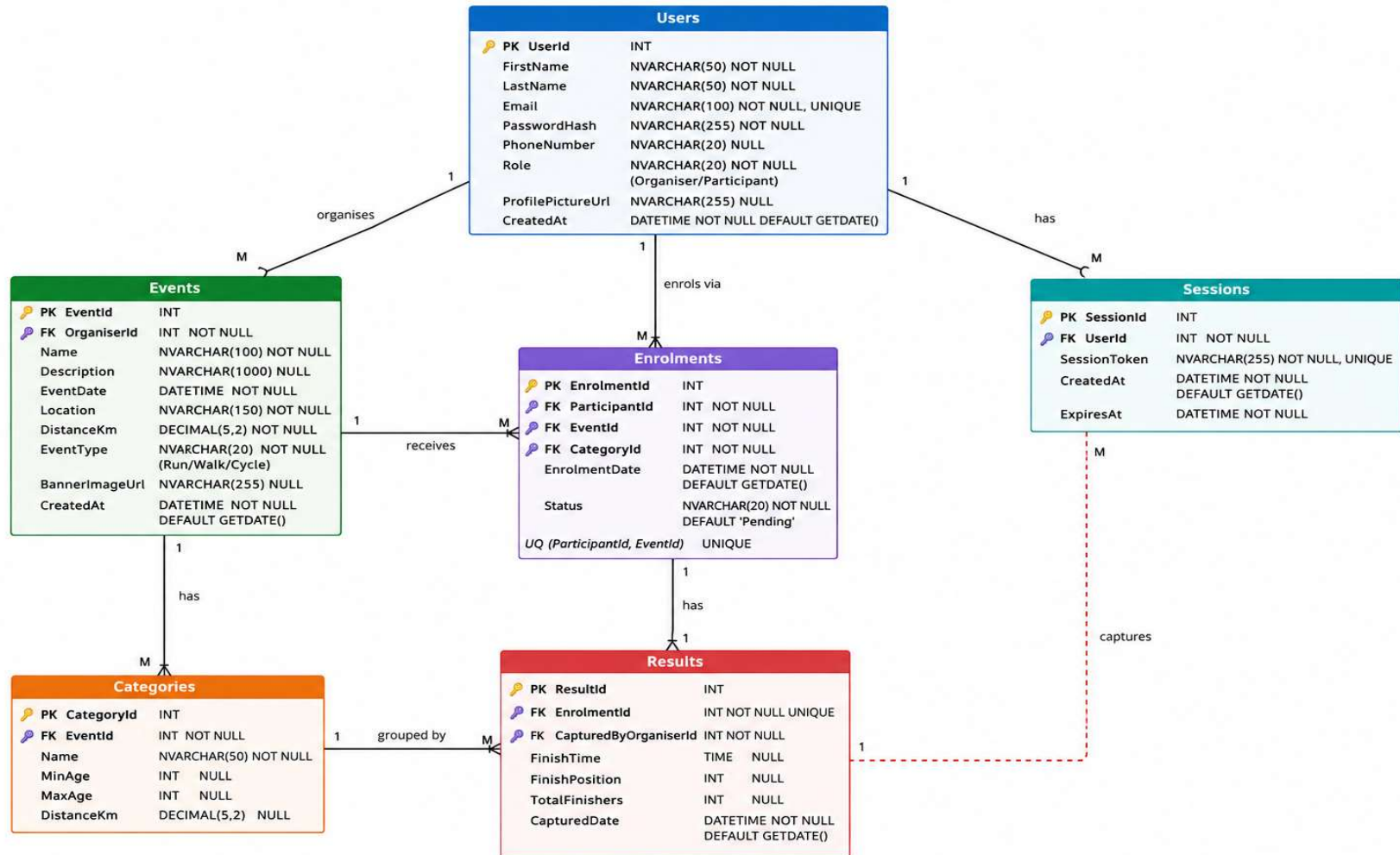


Section A - Entity Relationship Diagram (ERD)



RaceDay

API Endpoint Plan

Part 1, Section B

REST API Design Specification

1. Overview

This document lists every endpoint that the RaceDay REST API will expose in Part 2. The endpoint plan follows directly from the Entity Relationship Diagram (ERD) in Section A, which contains Users, divided into Organisers and Participants, as well as Events, Categories, Enrolments, and Results.

Roles Legend

Role	Meaning
None	Public endpoint; no login is required.
Any	Any authenticated user.
Organiser	Access is restricted to Organisers.
Participant	Access is restricted to Participants.

2. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Creates a new user account as either an Organiser or a Participant. The user's password is hashed before it is stored.	Public	firstName, lastName, email, password, role, phoneNumber	201 Created - new user account created successfully, without returning the password. 400 Bad Request - validation failed. 409 Conflict - email address is already registered.
POST	/api/auth/login	Validates the user's credentials and starts a session. The session stores the user's ID and role.	Public	email, password	200 OK - user summary and role returned successfully. 401 Unauthorized - invalid credentials.
POST	/api/auth/logout	Ends the current user's session.	Any	None. The authenticated User ID is obtained from the session.	200 OK – user has been successfully logged out.

3. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the profile information of the currently logged-in user, including the appropriate fields for either an Organiser or Participant.	Any	The authenticated User ID is obtained from the session	200 OK - profile object returned successfully. 401 Unauthorized - user is not authenticated.
PUT	/api/users/me	Updates the profile information belonging to the currently logged-in user.	Any	User ID (obtained from the session) firstName, lastName, phoneNumber, and role-specific fields	200 OK - updated profile returned successfully. 400 Bad Request - invalid information was submitted. 401 Unauthorized - user is not authenticated.
PUT	/api/users/me/profile-picture	Uploads or updates the Participant's profile picture.	Participant	User ID (obtained from the session) multipart/form-data containing an image file	200 OK - image URL returned successfully. 400 Bad Request - invalid image or request.

4. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists all available events. Optional filtering can be applied using date, location, and event type. This endpoint allows users to browse available events.	Public	None	200 OK - array containing the available events.
GET	/api/events/{id}	Returns the complete details of a specific event, including its associated categories.	Public	Event ID (id route parameter)	200 OK - event object returned successfully. 404 Not Found - requested event does not exist.
POST	/api/events	Creates a new event owned by the currently logged-in Organiser.	Organiser	Organiser ID (obtained from the authenticated session) name, description, eventDate, location, distanceKm, eventType	201 Created - new event created successfully. 400 Bad Request - invalid event information was submitted.

PUT	/api/events/{id}	Updates an existing event. Only the Organiser who owns the event is permitted to edit it.	Organiser	Event ID (id route parameter). Organiser ID is obtained from the authenticated session. name, description, eventDate, location, distanceKm, eventType	200 OK - event updated successfully. 403 Forbidden - Organiser does not own the event. 404 Not Found - requested event does not exist.
DELETE	/api/events/{id}	Deletes an event owned by the currently logged-in Organiser.	Organiser	Event ID (id route parameter). Organiser ID is obtained from the authenticated session	204 No Content - event deleted successfully. 403 Forbidden - Organiser does not own the event. 404 Not Found - requested event does not exist.
POST	/api/events/{id}/banner	Uploads a banner image for an event to Azure Blob Storage.	Organiser	Event ID (id route parameter). Organiser ID is obtained from the authenticated session multipart/form-data containing an image file	200 OK - image URL returned successfully. 403 Forbidden - Organiser does not own the event. 404 Not Found - requested event does not exist.

5. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Lists all categories available for a specific event.	Public	Event ID (eventId route parameter).	200 OK - array containing the event categories. 404 Not Found - requested event does not exist.
POST	/api/events/{eventId}/categories	Adds a new age or distance category to an event owned by the Organiser.	Organiser	Event ID (eventId route parameter), Organiser ID (obtained from the authenticated session), categoryName, minAge, maxAge, distanceKm	201 Created - new category created successfully. 403 Forbidden - Organiser does not own the event. 404 Not Found - requested event does not exist.
PUT	/api/categories/{id}	Updates a category belonging to an event owned by the Organiser.	Organiser	Category ID (id route parameter), Organiser ID (obtained from the authenticated session), categoryName, minAge, maxAge, distanceKm	200 OK - category updated successfully. 403 Forbidden - Organiser does not have permission to modify the category. 404 Not Found - requested category does not exist.

DELETE	/api/categories/{id}	Deletes a category from an event owned by the Organiser.	Organiser	Category ID (id route parameter), Organiser ID (obtained from the authenticated session)	204 No Content - category deleted successfully. 403 Forbidden - Organiser does not have permission to delete the category. 404 Not Found - requested category does not exist.
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6. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/enrolments	Enrols the currently logged-in Participant into an event under the selected category.	Participant	Event ID (eventId route parameter), Participant ID (obtained from the authenticated session) & categoryId	201 Created - enrolment record created successfully. 400 Bad Request - invalid enrolment information. 404 Not Found - specified event or category does not exist. 409 Conflict - Participant is already enrolled in the event.
GET	/api/enrolments/mine	Returns all enrolments belonging to the currently logged-in Participant, including their enrolment status.	Participant	Participant ID (obtained from the authenticated session)	200 OK - array containing the Participant's enrolments.
GET	/api/events/{eventId}/enrolments	Returns all Participants enrolled in a specific event owned by the Organisation.	Organisation	Event ID (eventId route parameter), Organisation ID (obtained from the authenticated session)	200 OK - array containing the event enrolments. 403 Forbidden - Organisation does not own the event.

					404 Not Found - requested event does not exist.
DELETE	/api/enrolments/{id}	Cancels the currently logged-in Participant's own enrolment.	Participant	Enrolment ID (enrolmentId route parameter), Participant ID (obtained from the authenticated session)	204 No Content - enrolment cancelled successfully. 403 Forbidden - Participant is not permitted to cancel the specified enrolment. 404 Not Found - enrolment does not exist.

7. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/result	Captures and publishes the finish time and finishing position for a Participant's enrolment after the Organiser's event has concluded.	Organiser	Result ID (id route parameter), Organiser ID (obtained from the authenticated session). finishTimeSeconds, finishPosition, totalFinishers	201 Created - result record created successfully. 400 Bad Request - invalid result information. 403 Forbidden - Organiser does not own the event. 404 Not Found - specified enrolment or related resource does not exist.
PUT	/api/results/{id}	Updates a previously captured result.	Organiser	finishTimeSeconds, finishPosition, totalFinishers	200 OK - result updated successfully. 403 Forbidden - Organiser does not have permission to update the result.

					404 Not Found - requested result does not exist.
GET	/api/results/mine	Returns the complete personal race history of the currently logged-in Participant across all completed events.	Participant	Participant ID (obtained from the authenticated session)	200 OK — array containing the Participant's results, including event name, event date, category, finishing time, and finishing position.
GET	/api/events/{eventId}/results	Returns all published results for an event owned by the Organiser.	Organiser	Event ID (eventId route parameter), Organiser ID (obtained from the authenticated session)	200 OK - array containing the event results. 403 Forbidden - Organiser does not own the event. 404 Not Found - requested event does not exist.

8. Organiser Dashboard

Additional Endpoint — Not Included in the Minimum List

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/organisers/dashboard	Returns a summary for the currently logged-in Organiser. The summary includes all events created by the Organiser, the total number of enrolments for each event, and upcoming event dates. This allows the application to retrieve the dashboard information through a single API call instead of assembling it from several separate endpoints.	Organiser	Organiser ID (obtained from the authenticated session)	200 OK - array containing eventId, eventName, eventDate, and totalEnrolments for each event.

9. Notes on Design Decisions

9.1 Nested Routes

Routes are nested under their parent resource when the child resource cannot logically exist without the parent resource. For example, `/api/events/{eventId}/categories` and `/api/events/{eventId}/enrolments`. This approach keeps the API design consistent with the one-to-many (1:M) relationships represented in the ERD.

9.2 Ownership and Role Checks

Organiser and Participant ownership checks are planned for every write endpoint that interacts with another user's data. A 403 Forbidden response will be returned when an authenticated user does not have permission to access or modify the relevant resource. This supports the role-based access requirements that will be implemented in Part 2.

9.3 Image Upload Endpoints

Image upload functionality is separated from the main create and update endpoints. This allows Azure Blob Storage, which will be added in Part 3, to be integrated without requiring changes to the core resource endpoints.